

Week 2 – Application Layer

The **Application Layer** is the top layer of the network model. It is the layer closest to the user.

When you use:

Chrome browser, Gmail, WhatsApp Web, YouTube, Google Drive, banking website, FTP, DNS lookup

you are using **application layer protocols**.

The application layer does not physically send data through cables or Wi-Fi. Instead, it provides rules for applications to communicate over the network.

Example:

When you open:

`www.google.com`

your browser uses:

DNS to find the IP address

HTTP/HTTPS to request the webpage

TCP/IP to transport the data

2.1 Principles of Network Applications

A **network application** is software that communicates with another software over a network.

Examples:

Browser communicates with web server

Email app communicates with mail server

WhatsApp communicates with WhatsApp server

Netflix app communicates with video server

Basic Idea

There are usually two sides:

Client → Request → Server
Client ← Response ← Server

Client

The client is the device or application that requests service.

Examples:

Your browser

Gmail app

Mobile banking app

YouTube app

Server

The server provides the service.

Examples:

Google web server

Gmail mail server

Netflix video server

Bank server

Example

When you type:

`https://www.amazon.in`

Your browser is the **client**.

Amazon's system is the **server**.

Your browser sends a request:

Please send me the Amazon homepage

Amazon server replies:

Here is the webpage

Types of Network Application Architecture

1. Client-Server Architecture

This is the most common model.

```
Client 1 \
Client 2  → Server
Client 3 /
```

Example:

Browser → Google server

Gmail app → Gmail server

ATM machine → Bank server

Advantages:

Centralized control

Easy to maintain

Secure management

Disadvantage:

If server fails, service may stop

2. Peer-to-Peer Architecture

In peer-to-peer, devices communicate directly with each other.

```
Peer 1 ↔ Peer 2 ↔ Peer 3
```

Example:

Torrent sharing

Some file sharing systems

Blockchain networks

Advantages:

No single central server needed

Load is distributed

Disadvantages:

Harder to control

Security risks are higher

2.2 Web and HTTP

The **Web** means websites and webpages that we access using browsers.

Examples:

Google

YouTube

Facebook

College website

Online banking website

The web mainly uses **HTTP** and **HTTPS**.

What is HTTP?

HTTP means:

HyperText Transfer Protocol

It is an application layer protocol used for communication between browser and web server.

Example:

```
Browser → HTTP Request → Web Server  
Browser ← HTTP Response ← Web Server
```

HTTP Request Example

When you open a website, your browser sends a request like:

```
GET /index.html HTTP/1.1  
Host: example.com
```

Meaning:

Please give me the index.html page from example.com

HTTP Response Example

The server replies:

```
HTTP/1.1 200 OK
Content-Type: text/html
```

Meaning:

Request successful. Here is the webpage.

Common HTTP Methods

GET

Used to request data.

Example:

Opening a webpage.

```
GET /home
```

POST

Used to send data.

Example:

Login form.

```
POST /login
```

PUT

Used to update existing data.

Example:

Updating profile information.

DELETE

Used to delete data.

Example:

Deleting a user account or record.

HTTP Status Codes

200 OK

Request successful.

301/302 Redirect

Page moved to another URL.

400 Bad Request

Client sent wrong request.

401 Unauthorized

Login required.

403 Forbidden

Access denied.

404 Not Found

Page does not exist.

500 Internal Server Error

Server has a problem.

HTTP vs HTTPS

HTTP

Data is sent in plain text.

Example:

```
username=admin  
password=12345
```

If attacker captures traffic, they may read it.

HTTPS

HTTPS means:

```
HTTP + SSL/TLS encryption
```

Data is encrypted.

Example:

```
a8f9x92ksl39sks...
```

Even if attacker captures traffic, they cannot easily read it.

2.3 Electronic Mail Protocols

Email works using different protocols.

Main email protocols:

SMTP

POP3

IMAP

Basic Email Flow

```
Sender Mail App → SMTP Server → Receiver Mail Server → Receiver Mail App
```

Example:

You send email from Gmail to Yahoo.

```
Your Gmail app → Gmail SMTP server → Yahoo mail server → Yahoo inbox
```

SMTP

SMTP means:

Simple Mail Transfer Protocol

SMTP is used to **send email**.

Common ports:

25 - SMTP
587 - SMTP with STARTTLS
465 - SMTP over SSL

Example:

When you click **Send** in Gmail, SMTP is used.

SMTP is like a postman who carries the letter from sender to receiver.

POP3

POP3 means:

Post Office Protocol version 3

POP3 is used to **download email** from server to device.

Common port:

110 - POP3
995 - POP3 over SSL

In POP3, emails are usually downloaded to one device.

Example:

You configure Outlook using POP3. Emails come to your laptop. They may be removed from the server depending on settings.

Best for:

Single device email access

Offline email storage

IMAP

IMAP means:

Internet Message Access Protocol

IMAP is used to access emails while keeping them on the server.

Common port:

143 - IMAP
993 - IMAP over SSL

Example:

You open Gmail on mobile and laptop. Same emails are visible on both devices.

Best for:

Multiple devices

Cloud-based email access

Synchronization

SMTP vs POP3 vs IMAP

Protocol	Purpose	Example
SMTP	Sends email	Sending Gmail message
POP3	Downloads email	Outlook downloads mail
IMAP	Syncs email	Gmail on phone and laptop

Simple memory trick:

SMTP = Send Mail
POP3 = Pull mail to one device
IMAP = Internet mail sync

2.4 DNS

DNS means:

Domain Name System

DNS converts domain names into IP addresses.

Humans remember names:

www.google.com

Computers communicate using IP addresses:

142.250.183.36

DNS works like the phonebook of the internet.

Why DNS is Needed

Imagine remembering every website by IP address:

142.250.183.36
157.240.239.35
13.107.42.14

That is difficult.

So DNS allows us to use names:

google.com
facebook.com
microsoft.com

DNS Working Example

When you type:

www.google.com

Steps:

1. Browser checks local cache.
2. Operating system checks DNS cache.
3. Query goes to DNS resolver.
4. Resolver asks root DNS server.
5. Root points to .com server.
6. .com server points to Google DNS server.
7. Google DNS gives IP address.
8. Browser connects to that IP.

Simple flow:

www.google.com → DNS → IP address → Web server

Important DNS Records

A Record

Maps domain to IPv4 address.

`example.com → 93.184.216.34`

AAAA Record

Maps domain to IPv6 address.

CNAME Record

Alias name.

Example:

`www.example.com → example.com`

MX Record

Mail server record.

Example:

`gmail.com mail handled by Google mail servers`

TXT Record

Stores text information.

Used for:

SPF

DKIM

Domain verification

Email security

Practical DNS Commands

In Windows:

`nslookup google.com`

In Linux:

```
dig google.com  
ping google.com
```

Example output:

```
google.com has address 142.250.183.36
```

DNS Security Issues

DNS Spoofing

Attacker gives fake IP address.

Example:

You type:

```
bank.com
```

But attacker redirects you to fake bank website.

DNS Cache Poisoning

Fake DNS record is stored in cache.

DNS Tunneling

Attackers use DNS queries to secretly send data.

Protection

Use HTTPS

Use DNSSEC

Use trusted DNS servers

Monitor DNS logs

2.5 Content Distribution Networks

CDN means:

Content Distribution Network

A CDN is a group of servers placed in different locations to deliver content faster.

Examples:

Cloudflare

Akamai

Amazon CloudFront

Google CDN

Azure CDN

Why CDN is Needed

Suppose a website server is in the USA.

A user from India accesses it.

Without CDN:

India user → USA server

This may be slow.

With CDN:

India user → Nearby CDN server in India/Singapore

This is faster.

CDN Example

You watch a YouTube video.

The video may not come directly from the main Google server in the USA.

It may come from a nearby CDN server.

This reduces:

Delay

Buffering

Bandwidth load

Server pressure

How CDN Works

1. Website stores static content in CDN.
2. User requests website.
3. DNS directs user to nearest CDN server.
4. CDN server sends cached content.
5. If content is not available, CDN fetches it from origin server.

Simple flow:

User → Nearby CDN → Origin Server

What Content CDN Delivers

Images

Videos

CSS files

JavaScript files

Software downloads

Documents

Web pages

Example:

logo.png
style.css
script.js
video.mp4

Benefits of CDN

Faster website loading

Less load on main server

Better user experience

Protection from DDoS attacks

High availability

Global performance improvement

CDN Security Benefits

CDNs can provide:

DDoS protection

Web Application Firewall

TLS/SSL support

Bot protection

Rate limiting

Caching protection

Complete Real-Life Example

When you open:

`https://www.youtube.com`

Many application layer concepts work together.

Step 1: DNS

Your system asks:

What is the IP address of youtube.com?

DNS replies with IP address.

Step 2: HTTPS

Browser creates secure encrypted connection.

Step 3: HTTP Request

Browser sends:

GET /homepage

Step 4: Server Response

YouTube sends webpage content.

Step 5: CDN

Videos, thumbnails, and images may come from nearby CDN servers.

Step 6: Application Layer

Your browser displays the website.

Simple Summary

Topic	Meaning
Network Applications	Apps communicating over network
Client-Server	Client requests, server responds
HTTP	Protocol for websites
HTTPS	Secure HTTP
SMTP	Sends email
POP3	Downloads email
IMAP	Syncs email
DNS	Converts name to IP
CDN	Delivers content from nearest server

Application layer is the layer where users directly interact with network services like web, email, DNS, and content delivery.

Computer Networking: A Top-Down Approach by Kurose & Ross, and Chapter 2: Application Layer, plus CDN from Chapter 7, Section 7.2.4.

Week 2: Application Layer

The **Application Layer** is the top layer of the Internet protocol stack. It is where user-facing applications work.

Examples:

```
Browser
Email
YouTube
WhatsApp Web
DNS
File transfer
Video streaming
```

In the textbook, Chapter 2 covers:

```
2.1 Principles of Network Applications
2.2 Web and HTTP
2.4 Electronic Mail
2.5 DNS
```

And CDN is covered later under multimedia networking, especially streaming video.

2.1 Principles of Network Applications

A **network application** is a program that runs on different end systems and communicates over the network.

Example:

```
Your browser → Google server
Your Gmail app → Gmail mail server
Your YouTube app → YouTube CDN/server
```

Client and Server

The **client** requests service.

The **server** provides service.

Example:

```
Client: Your Chrome browser
Server: www.google.com server
```

Flow:

```
Browser sends request
Server sends response
```

Example:

User types: `www.amazon.in`
Browser asks Amazon server for homepage
Amazon server sends HTML, CSS, images, scripts

Client-Server Architecture

In this model, one central server provides service to many clients.

```
Client 1 \
Client 2  → Server
Client 3 /
```

Examples:

Banking website
Gmail
Amazon
College portal

Peer-to-Peer Architecture

In P2P, systems communicate directly with each other.

```
Peer ↔ Peer ↔ Peer
```

Examples:

Torrent
Blockchain
Some file-sharing systems

2.2 Web and HTTP

The **Web** is a collection of webpages, websites, images, videos, and resources accessed through browsers.

The protocol used by the Web is **HTTP**.

HTTP means:

HyperText Transfer Protocol

Basic HTTP Flow

```
Browser → HTTP Request → Web Server
Browser ← HTTP Response ← Web Server
```

Example:

When you type:

```
http://example.com/index.html
```

Browser sends:

```
GET /index.html HTTP/1.1  
Host: example.com
```

Server replies:

```
HTTP/1.1 200 OK  
Content-Type: text/html
```

Meaning:

Request successful. Here is the webpage.

HTTP Methods

Method	Use
GET	Request data
POST	Send data
PUT	Update data
DELETE	Delete data

Example:

```
Opening a webpage → GET  
Submitting login form → POST  
Updating profile → PUT  
Deleting account → DELETE
```

HTTP Status Codes

Code	Meaning
200	OK
301/302	Redirect
400	Bad request
401	Login required
403	Forbidden
404	Page not found
500	Server error

HTTP vs HTTPS

HTTP sends data in plain text.

HTTPS uses encryption.

Example:

HTTP:
username=admin&password=12345

HTTPS:
Encrypted unreadable data

So for banking, Gmail, shopping, and login pages, HTTPS is essential.

2.3 Electronic Mail Protocols: SMTP, POP3, IMAP

Email does not use only one protocol. It uses different protocols for sending and receiving.

Email Flow

Sender Mail App → Sender Mail Server → Receiver Mail Server → Receiver Mail App

Example:

Gmail user sends mail to Yahoo user

Flow:

Gmail app → Gmail SMTP server → Yahoo mail server → Yahoo inbox

SMTP

SMTP means:

Simple Mail Transfer Protocol

SMTP is used to **send email**.

Example:

When you click **Send** in Gmail, SMTP is used.

Common ports:

25 SMTP
587 SMTP with STARTTLS
465 SMTP over SSL

Simple meaning:

SMTP = sending mail

POP3

POP3 means:

Post Office Protocol version 3

POP3 is used to download email from server to one device.

Example:

Outlook downloads mail from server to laptop

Common ports:

110 POP3
995 POP3 over SSL

POP3 is useful when:

You mainly use one device
You want offline email storage

IMAP

IMAP means:

Internet Message Access Protocol

IMAP keeps emails on the server and syncs them across devices.

Example:

You read Gmail on mobile
Same mail shows as read on laptop also

Common ports:

143 IMAP
993 IMAP over SSL

Simple comparison:

Protocol	Purpose
SMTP	Send email
POP3	Download email
IMAP	Sync email across devices

Memory trick:

SMTP = Send
POP3 = Pull to one device
IMAP = Internet mail sync

2.4 DNS

DNS means:

Domain Name System

DNS converts website names into IP addresses.

Humans remember:

`www.google.com`

Computers use:

`142.250.xxx.xxx`

So DNS is like the **phonebook of the Internet**.

DNS Example

When you type:

`www.youtube.com`

Steps:

1. Browser checks cache
2. OS checks DNS cache
3. DNS resolver is contacted
4. Root DNS server is contacted
5. .com DNS server is contacted
6. YouTube authoritative DNS gives IP
7. Browser connects to that IP

Simple flow:

Domain name → DNS → IP address → Server

Important DNS Records

Record	Purpose
A	Domain to IPv4
AAAA	Domain to IPv6
CNAME	Alias name
MX	Mail server
TXT	Verification/security records

Example:

google.com → A record → IP address
gmail.com → MX record → mail server

Practical Commands

Windows:

```
nslookup google.com
```

Linux:

```
dig google.com  
ping google.com
```

2.5 Content Distribution Networks

CDN means:

Content Distribution Network

A CDN is a group of servers placed in different geographical locations to deliver content faster.

The textbook explains that large video companies cannot depend on one single data center because users are spread across the world, paths may cross many links and ISPs, popular videos may be sent repeatedly, and a single data center can become a failure point. CDNs solve this by storing copies of videos and web content in multiple locations and directing users to a nearby/better server.

Examples:

Cloudflare
Akamai
Amazon CloudFront
Google CDN
Netflix CDN

Without CDN

User in India → Server in USA

This may cause:

High delay
Buffering
Slow page loading
Heavy server load

With CDN

User in India → Nearby CDN server in India/Singapore

Benefits:

Fast loading
Less buffering
Reduced main server load
Better availability
DDoS protection

Example:

When you watch YouTube, the video may come from a nearby Google CDN server, not directly from one central YouTube server.

Full Example: Opening YouTube

When you open:

`https://www.youtube.com`

Many application-layer concepts work together:

1. DNS finds YouTube IP address
2. HTTPS creates secure connection
3. Browser sends HTTP request
4. YouTube server/CDN sends response
5. CDN delivers videos/images faster
6. Browser displays the webpage

Simple final flow:

User → Browser → DNS → HTTP/HTTPS → Server/CDN → Webpage/Video

Final Summary

Topic	Simple Meaning
Network Applications	Programs communicating over network
HTTP	Web communication protocol
SMTP	Sends email
POP3	Downloads email
IMAP	Syncs email
DNS	Converts name to IP
CDN	Delivers content from nearby servers

The **Application Layer** is where real user services exist: web browsing, email, DNS lookup, file transfer, and video streaming.

Chapter Mapping (Application Layer)

All your topics come from:

Chapter 2 - Application Layer
Chapter 7 - CDN (Multimedia Networking)

2.1 Principles of Network Applications

Textbook Reference:

- **Chapter 2 start**
- Approx pages: ~**Page 36 to Page 80** (PDF ~64–110)

What to read:

- Application architectures (Client-Server, P2P)
- Process communication
- Sockets
- Transport services

This section builds the **foundation of all application-layer protocols**.

2.2 Web and HTTP

Textbook Reference:

- **Section 2.2 Web and HTTP**

- Approx pages: ~**Page 80 to Page 115 (PDF ~110–145)**

What to read:

- HTTP request/response format
- Persistent vs non-persistent connections
- Cookies
- Web caching
- Proxy servers

This is the **most important exam + practical topic**.

2.3 Electronic Mail (SMTP, POP3, IMAP)

Textbook Reference:

- **Section 2.4 Electronic Mail**
- Approx pages: ~**Page 115 to Page 130 (PDF ~145–160)**

Key proof from your PDF:

- SMTP explanation starts at:
Page 121 (PDF 149)

What to read:

- Email architecture
- SMTP working (steps of sending mail)
- Mail servers and queues
- POP3 and IMAP differences

Example explained in book:

Alice → Bob email flow using SMTP

2.4 DNS

Textbook Reference:

- **Section 2.5 DNS**
- Approx pages: ~**Page 130 to Page 150 (PDF ~160–170)**
- DNS explanation:
Page 133 (PDF 161)

- DNS message format:
Page 140–141 (PDF 168–169)

What to read:

- DNS hierarchy (root, TLD, authoritative)
- DNS query process
- DNS records (A, MX, CNAME)
- DNS message structure

Important concept:

DNS is a distributed database (NOT centralized)

2.5 Content Distribution Networks (CDN)

Textbook Reference:

- **Chapter 7 – Section 7.2.4**
- Approx pages: ~**Page 600+** (PDF ~630+)
- CDN working:
Page 604 (PDF 632)
- CDN architecture (Enter-deep vs Bring-home)
- Content replication
- DNS-based redirection
- Streaming optimization

Important real-world concept:

DNS is used to redirect users to nearest CDN server

Final Quick Reference Table

Topic	Section	Book Page	PDF Page
Principles	2.1	~36–80	~64–110
HTTP	2.2	~80–115	~110–145
Email	2.4	~115–130	~145–160
DNS	2.5	~130–150	~160–170
CDN	7.2.4	~600+	~630+